Oil & FX

Pricing a Brent Option in Euros (Monte Carlo, Black–Scholes)

Bilal Ashraf — MSc Mathematical Engineering, Université Côte d'Azur

1. Question

A European airline or refiner buys its oil in US dollars but earns euros. It carries two risks at once: the oil price and the dollar. To protect itself against a rise of its oil bill *in euros*, it can buy a call option on the Brent price expressed in euros.

How much does this option cost, and how does the correlation between oil and the dollar change its price?

2. Definitions

FX (foreign exchange). The currency market. EUR/USD = 1.10 means that one euro is worth 1.10 dollars. I write $Z = 1/\text{EURUSD}$ for the value of one dollar in euros.

Brent. The reference price of crude oil in Europe, quoted in dollars per barrel.

Call option. The right, not the obligation, to buy an asset at a fixed strike K at maturity T. Its payoff is $\max(S_T - K, 0)$.

Black–Scholes. Closed-form price of a European option when the asset follows a geometric Brownian motion (GBM), i.e. has lognormal returns with constant volatility.

Monte Carlo. Pricing by simulation: draw many scenarios, average the discounted payoff.

Quanto adjustment. Correction of the drift of an asset quoted in a foreign currency when the payoff is valued in the domestic currency.

3. Method

Model. The Brent price in dollars S and the dollar value in euros Z are two correlated GBMs, with volatilities σ_S, σ_Z and correlation ρ. The Brent price in euros is their product:

$$Y_t = S_t\, Z_t.$$

Dynamics under the euro pricing measure. The dollar, seen from Europe, is an asset that earns the dollar rate, so Z drifts at $r_{\text{EUR}} - r_{\text{USD}}$. The product Y is an asset priced in euros and must drift at r_{EUR}. This forces the drift of S:

$$\frac{dZ}{Z} = (r_{\text{EUR}} - r_{\text{USD}})\,dt + \sigma_Z\, dW^Z, \qquad \frac{dS}{S} = (r_{\text{USD}} - \rho\,\sigma_S\sigma_Z)\,dt + \sigma_S\, dW^S.$$

The term $-\rho\,\sigma_S\sigma_Z$ is the quanto adjustment. Without it, the simulated price would be biased.

Monte Carlo. I draw 200,000 pairs of correlated Gaussian shocks ($z_2 = \rho z_1 + \sqrt{1-\rho^2}\,\varepsilon$), compute S_T, Z_T and the discounted payoff $e^{-r_{\text{EUR}}T}\max(S_T Z_T - K, 0)$.

Closed-form check. The product of two lognormal variables is lognormal. Y is therefore a GBM with drift r_{EUR} and volatility

$$\sigma_Y = \sqrt{\sigma_S^2 + \sigma_Z^2 + 2\,\rho\,\sigma_S\sigma_Z},$$

and the call is given by the Black–Scholes formula applied to $Y_0 = S_0 Z_0$ with σ_Y. The two methods must agree, which validates the simulation, including the quanto term.

4. Code

Full code: `brent_eur_option.py` (100 lines, NumPy, SciPy and Matplotlib).

```python
def monte_carlo_call(strike_eur, rho, rng):
    t = MATURITY
    z1 = rng.standard_normal(N_PATHS)
    z2 = rho * z1 + np.sqrt(1 - rho**2) * rng.standard_normal(N_PATHS)  # correlated

    drift_oil = R_USD - rho * SIGMA_OIL * SIGMA_FX                      # quanto adjustment
    oil = BRENT_USD * np.exp((drift_oil - 0.5 * SIGMA_OIL**2) * t
                             + SIGMA_OIL * np.sqrt(t) * z1)
    fx = (1 / EURUSD) * np.exp((R_EUR - R_USD - 0.5 * SIGMA_FX**2) * t
                               + SIGMA_FX * np.sqrt(t) * z2)

    payoff = np.exp(-R_EUR * t) * np.maximum(oil * fx - strike_eur, 0.0)
    return payoff.mean(), payoff.std(ddof=1) / np.sqrt(N_PATHS)

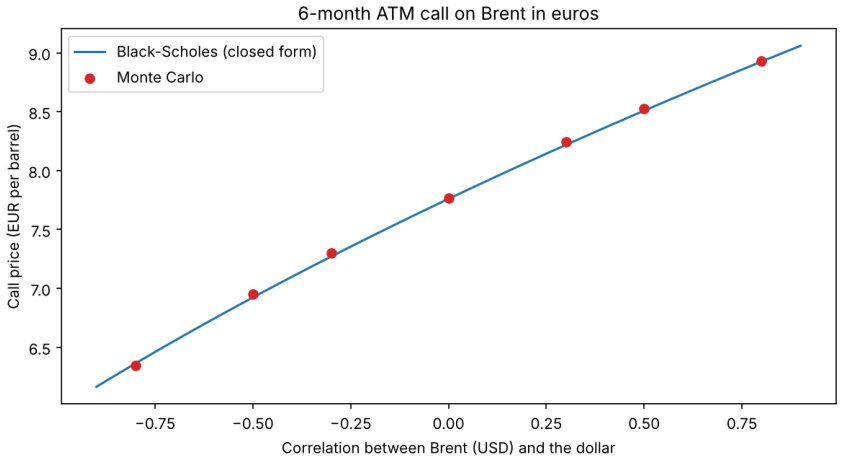
def sigma_eur(rho):
    return np.sqrt(SIGMA_OIL**2 + SIGMA_FX**2 + 2 * rho * SIGMA_OIL * SIGMA_FX)
```

5. Results

Parameters (illustrative, not estimated from market data): Brent at 80 USD, EUR/USD at 1.10, hence 72.73 EUR per barrel; $\sigma_S = 35\%$, $\sigma_Z = 8\%$, $r_{\text{USD}} = 4\%$, $r_{\text{EUR}} = 2.5\%$. The option is an at-the-money call (strike 72.73 EUR) with a 6-month maturity.

Validation. For $\rho = -0.30$, Monte Carlo gives 7.280 EUR (standard error 0.027) against 7.270 EUR with Black–Scholes: a difference of 0.13%, inside the Monte Carlo confidence interval.

Correlation ρ	Volatility in EUR	Monte Carlo (EUR)	Black–Scholes (EUR)	vs. $\rho = 0$
−0.8	29.00%	6.346	6.364	−18.0%
−0.5	31.76%	6.951	6.923	−10.8%
−0.3	33.48%	7.298	7.270	−6.3%
0.0	35.90%	7.768	7.760	–
+0.3	38.17%	8.244	8.218	+5.9%
+0.5	39.61%	8.528	8.508	+9.6%
+0.8	41.68%	8.932	8.925	+15.0%



Reading.

- The hedge costs about 10% of the barrel price for six months (7.27 EUR on 72.73 EUR).
- When oil rises while the dollar falls ($\rho < 0$), the two effects partly offset each other for a euro buyer: the Brent price in euros is less volatile than in dollars and the option is cheaper. The dollar then acts as a partial natural hedge.
- The price moves from −18% to +15% over the range of correlations tested. Ignoring the FX leg, or assuming independence, misprices the hedge.

6. Limitations and extensions

- Parameters are illustrative. The next step is to estimate the two volatilities and the correlation from daily returns of Brent futures and EUR/USD, and to study how stable the correlation is over time.
- Constant volatility and correlation: no volatility smile, no jumps, although oil prices jump on supply shocks.
- Oil is modelled as a spot asset without convenience yield. In practice the option is written on a futures contract and priced with the Black (1976) formula.
- Extension: an Asian (average-price) option, which is the standard hedge for airlines and has no closed form, so Monte Carlo becomes necessary.